



# EXPLOITS UNIVERSITY

BSc INFORMATION COMMUNICATION AND TECHNOLOGY

**BICT 3202 · OBJECT-ORIENTED ANALYSIS AND DESIGN**

## WEEK 7

**Assignment: Object-Oriented Analysis — Dynamic Modelling**

**PROGRAMME**

BSc Information Communication and Technology

**COURSE CODE / YEAR**

BICT 3202 · Year 3

**LECTURER**

Francis Fweta — ICT Lecturer

**DELIVERY**

2h Lecture · 1h Tutorial · 1h Practical · 10h Independent Learning

*Prepared by Francis Fweta · Exploits University · Week 7 of 16*



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## ASSIGNMENT INSTRUCTIONS

- Answer ALL questions in Sections A, B and C.
- Read the Week 7 module and your lecture notes before attempting each question.
- Draw state-machine diagrams neatly (by hand or using a UML/CASE tool) with correct notation.
- Use a filled circle for the initial state and a bull's-eye for the final state; label every transition.
- Justify guards and every transition you include.
- Submit by the date given by your lecturer. Total: 50 marks.

### Section A — Short-Answer Questions (20 marks)

#### Question A1

Define dynamic modelling and explain why it is needed in an object-oriented system. [2 marks]

#### Question A2

Define an event and give two examples from different categories. [2 marks]

#### Question A3

Define a state and differentiate it from an event. [3 marks]

#### Question A4

What is a transition? What is a guard condition? Give an example of each. [3 marks]

#### Question A5

Explain entry, exit and do behaviour, with one example of each. [3 marks]

#### Question A6

Define an operation and differentiate it from an event. [2 marks]

#### Question A7

What is concurrency? Differentiate sequential from concurrent behaviour. [3 marks]

#### Question A8

Explain why a class diagram alone is insufficient for modelling a complex Order object. [2 marks]

### Section B — Application Questions (20 marks)

#### Question B1 (12 marks)

'A bank account can be Active, Blocked or Closed. The events that occur are login, wrongPIN, blockAccount, unblockAccount and closeAccount.'

- (a) Identify the states and the events. [2 marks]
- (b) Draw the state-machine diagram with an initial state. [4 marks]
- (c) Add a guard that blocks the account after three wrong PIN entries. [2 marks]
- (d) Identify any transition that should NOT be allowed, and justify your answer. [4 marks]

### Question B2 (8 marks)

Consider an online payment with states Pending, Processing, Paid and Failed, and events submitPayment, paymentSuccessful and paymentFailed.

- (a) Draw the state-machine diagram. [3 marks]
- (b) Add guards to distinguish the successful and failed outcomes. [2 marks]
- (c) Add entry behaviour to the Paid state (e.g. send a confirmation email). [1 mark]
- (d) Explain, in words, what the diagram says as a story. [2 marks]

## Section C — Case Study (10 marks)

'A food delivery application performs the following after an order is placed: payment processing, restaurant order preparation, driver assignment and customer notification. An order moves from Placed to Preparing, then to Out for Delivery, then to Delivered (or Cancelled).'

As an object-oriented analyst:

- (a) Identify the states and events of the Order. [3 marks]
- (b) Draw the Order state-machine diagram. [3 marks]
- (c) Explain which activities could occur concurrently and why. [4 marks]

## Marking Scheme

| Section | Content                              | Marks |
|---------|--------------------------------------|-------|
| A       | Short-answer questions (8 questions) | 20    |
| B       | Application questions (12 + 8)       | 20    |
| C       | Case study — food delivery           | 10    |
|         | TOTAL                                | 50    |

## Submission & Academic Integrity

- Submit a typed document (or neatly handwritten, as directed by your lecturer).
- This assignment is individual work. Plagiarism or copying will attract a penalty under university regulations.
- State-machine diagrams must use correct UML notation — initial state, states, labelled transitions, guards, final state.



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- Cite any sources you consult using the format used in the Week 7 module references.